Segmentation

evaluation

Silhuette score

Score

```
In [107]: from sklearn.metrics import silhouette_score

In [108]: score = silhouette_score(X, labels, metric='euclidean')

In [109]: print(score)

0.7732699147412556
```

Clustering on sample_prepared / Models / KMeans (k=4)

Summary

VARIABLES

Variables importance

CLUSTERS

Heatmap

Cluster profiles

Scatter plot

Detailed metrics

Measure	Value
Silhouette	0.7633
Inertia	5.749

https://scikit-learn.org/stable/auto_examples/cluster/plot_kmeans_silhouette_analysis.html

Silhouette

The Silhoutte coefficient for a point:

- shows how close each point in one cluster is to points in other clusters

- has a value between -1 and 1

+1: members of one cluster are very far from members of another cluster

0: they are very close

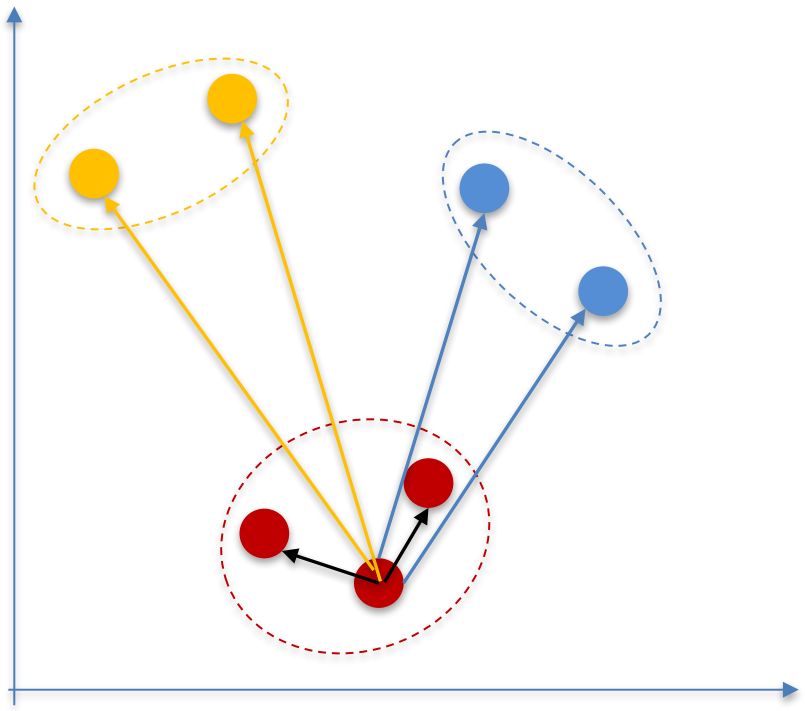
-1: they are most likely to be classified in the wrong cluster

$$S_i = \frac{b_i - a_i}{\max(a_i, b_i)}$$

a = average distance to other objects in the same cluster same cluster

b = The average distance to other objects in a different but closest cluster

$$S = avg\ (S_1, S_2, \dots, S_n)$$



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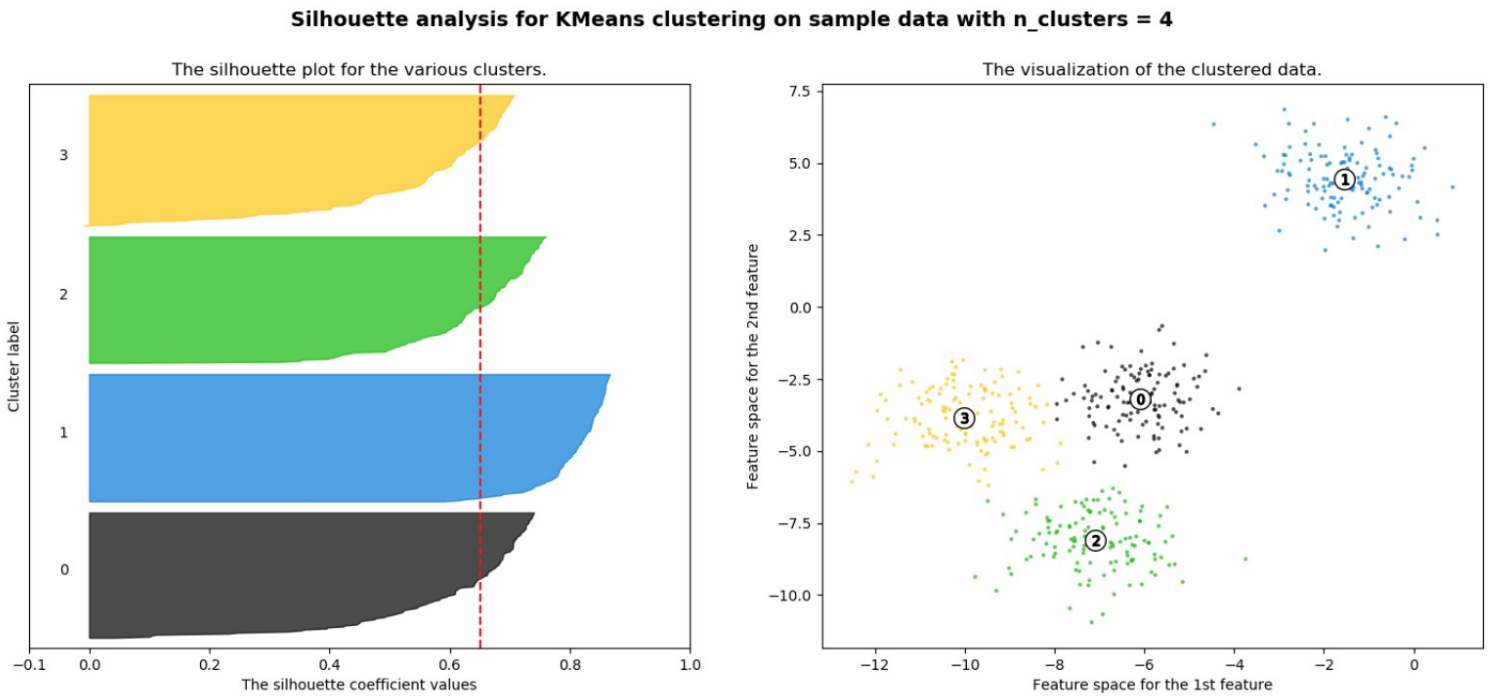

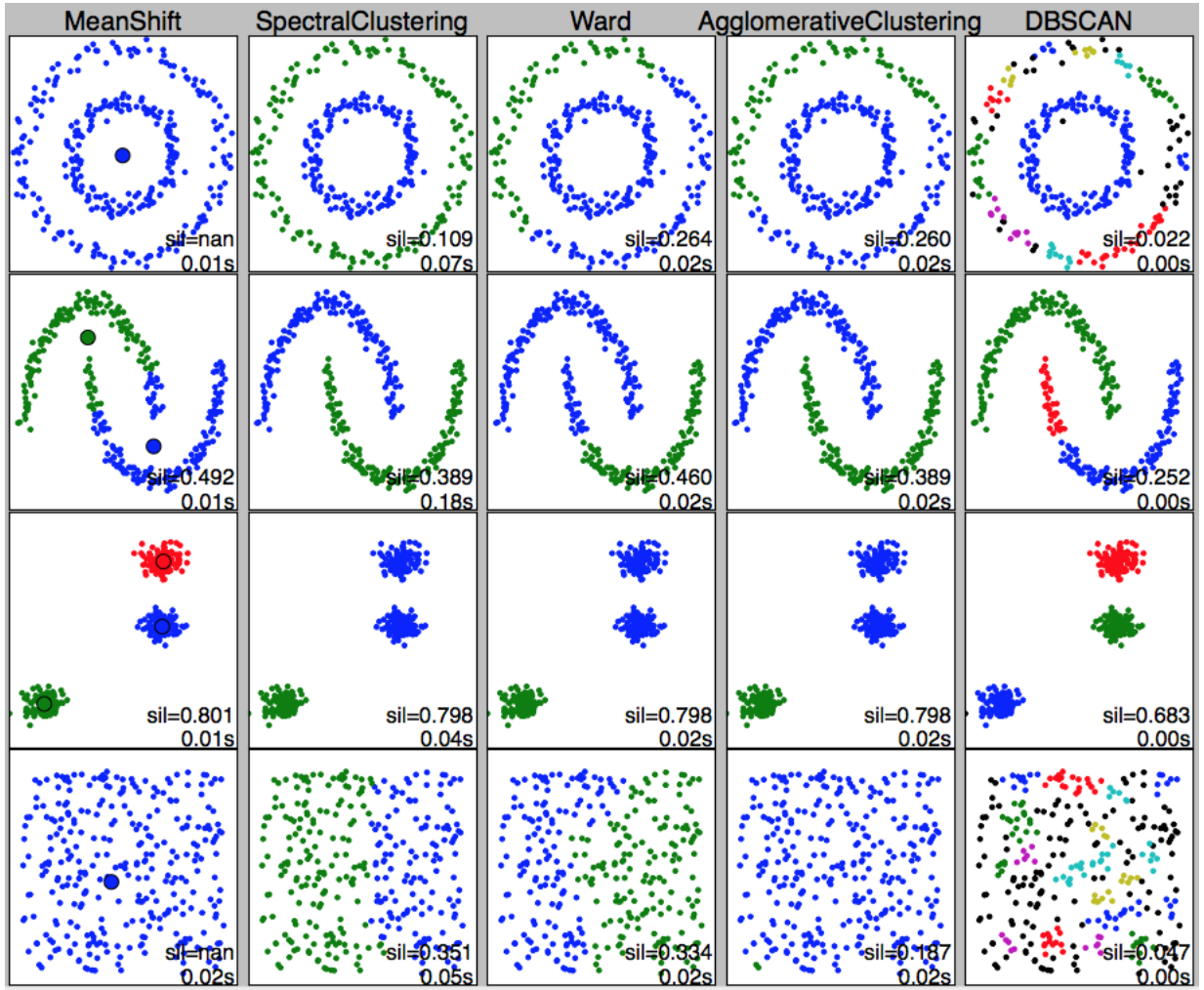

https://stackoverflow.com/questions/31762165/sklearn-silhouette-score-different-for-same-clustering